

Factors Influencing Error Propagation in the Measurement Process to Ensure Product Conformity

- Bryam Sebastián Girón Castro
- Juan Francisco Jiménez Faicán
- **Tutor:** Ing. Milton Barragán Landy, Mgt. PhD.

Abstract

The thesis integrates a systematic review, INEN data, surveys/case studies, and semantic embeddings to analyze error propagation and measurement uncertainty in Cuenca’s industry. This approach identifies gaps between the literature and practice and prioritizes strategies to strengthen the measurement system and improve conformity.

Objectives

General Objective

To determine the factors that influence the measurement process, in order to ensure product and service conformity and reduce error propagation, as an exploratory study in the context of the city of Cuenca, Ecuador.

Specific Objective

To propose strategies to mitigate error propagation in production systems, aimed at ensuring product or service conformity.

Identify the factors that influence the measurement process, and the methods currently used in industry to prevent error propagation in measuring equipment and ensure product or service conformity, through a systematic literature review.

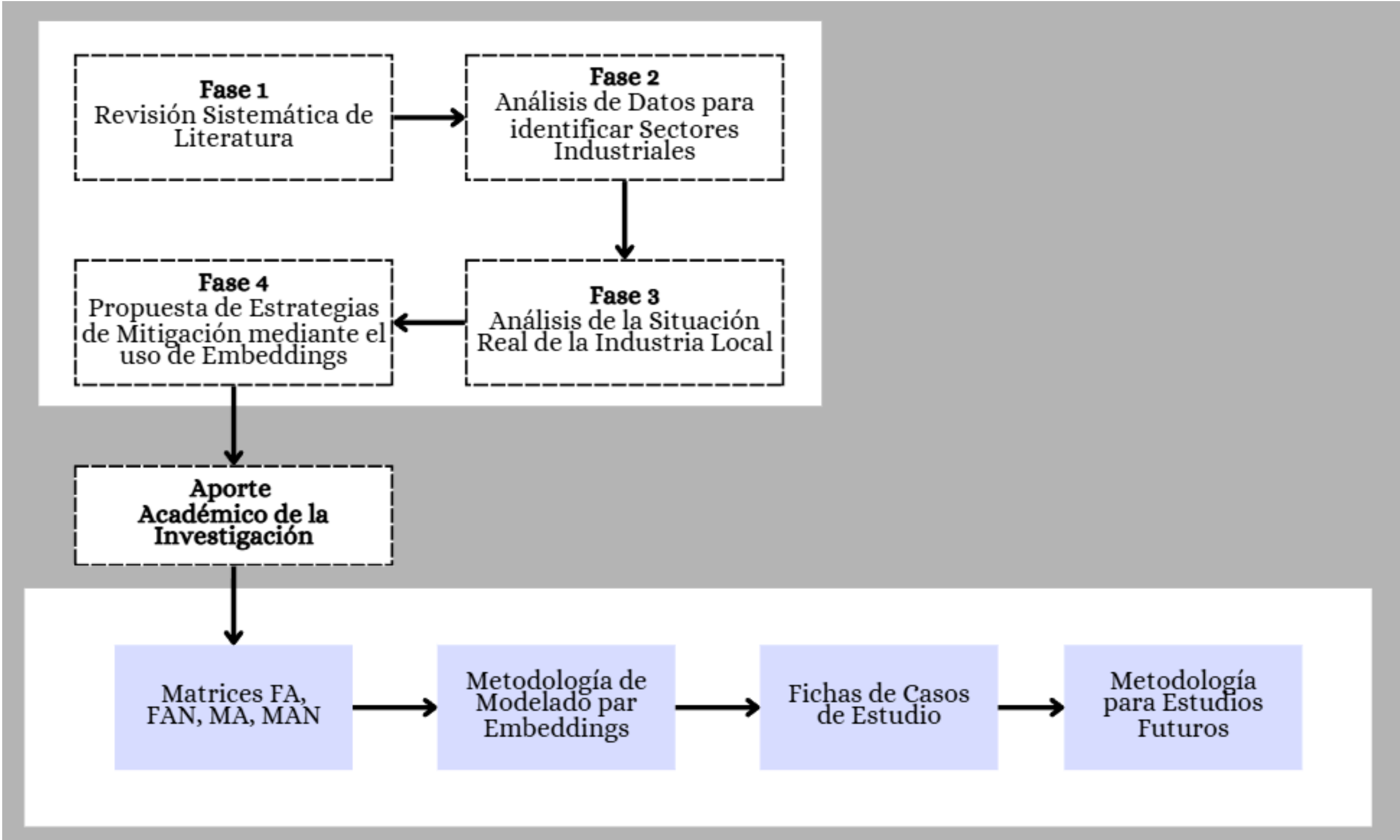
Specific Objective

Analyze the industrial sectors that are most vulnerable to error propagation affecting product or service conformity.

Evaluate the methods implemented in industry to prevent error propagation in measuring equipment that affects product or service conformity, in the case-study companies.

Specific Objective

Methodology



Curricular Integration

- Research Methodology
- Programming Languages
- Analytical Statistics
- Statistical Inference Techniques
- Quality Management
- Experimental Design
- Lean Services and Six Sigma

Applied Engineering

- Standards / guidelines (application in the study)**
- PRISMA 2020
 - ISO/IEC Guide 98-3 (GUM)
 - ISO/IEC 17025:2017 / ISO 9001:2015
 - ISIC Rev. 4
- Design constraints (real-world conditions considered)**
- Partial INEN data due to cutoff date
 - Prioritize low-cost, high-impact actions
 - Limited time availability
 - Confidential handling of sensitive information

Author Keywords

Error propagation

Conformity

Quality Control

Measurement process

Uncertainty

Metrology

Research Results

- The systematic review consolidated a framework of Grouped Factors (FA) and Grouped Methods (MA), in which a few elements account for the greatest recurrence.
- FA2 stands out as the most frequent factor and MA50 as the most dominant method, with strong relationships in FA–MA pairings.
- A pronounced geographic bias was also observed in the scientific output (greater presence of Europe and the U.S.).

- Calibration activity is concentrated in a few ISIC sectors, especially C (manufacturing), as well as G, H, and M, which guides sectoral prioritization.
- By magnitude, mass/scales predominates, followed by magnitudes such as length, volume/flow, and temperature, indicating where metrological exposure is greatest.
- Seasonality is observed in annual demand; in 2025 the data are truncated but maintain the 2023–2024 trend.

- The literature–industry integration using semantic embeddings showed that most FAN–FA and MAN–MA similarities fall within a medium gap, with few low-gap cases.
- Coverage between recommended methods and those used per case remained low (~16%–24%), evidencing a persistent practice gap.
- The clusters reveal a theoretical core of solutions alongside local variants, which enabled the construction of problem–method pathways to prioritize strategies.

Conclusions

Conclusion 1

Error propagation and measurement uncertainty are a systemic phenomenon, not an isolated failure of the operator or the instrument. The theory–practice relationship shows moderate correspondences between FA and FAN, with medium gaps of FAN→FA ≈ 0.3918. This implies interconnected weaknesses that reinforce one another and increase the system’s total uncertainty.

Conclusion 2

Metrological intensity does not guarantee greater methodological maturity in the sectors analyzed. Based on 1,046 INEN calibration items, the suboptimal component dominates ~72%–75%, while the aligned component remains at ~18%–19% and the potential at ~6%–9%. Therefore, the problem persists more due to the types of methods used than because of a lack of metrological activity.

Conclusion 3

There is a consistent gap between local methods (MAN) and those from the literature (MA), reflected in low coverage of recommendations. The highest medium gap reaches MAN→MA = 0.3991 (overall similarity ~0.6008). In practice, no case exceeds 25% coverage, with a maximum of ~24.24% and a minimum of 16.13%.

Conclusion 4

Although there are strong matches, the system is weighted toward suboptimal practices, limiting the closure of critical gaps. Targeted alignments are observed such as MAN1040–MA1 (0.7499) and MAN1022–MA55 (0.7106), demonstrating that effective methods are indeed available. The key improvement is to reconfigure the methodological portfolio toward aligned/potential practices that control uncertainty, traceability, and metrological stability.

References

- (Fang et al., 2005)
- (Jakubiec & Płowucha, 2013)
- (Mahmud et al., 2011)
- (Turley et al., 2014)
- (Wang et al., 2013)
- (Weckenmann et al., 1998)
- (Losada-Urzáiz et al., 2015)
- (Juran, 1988)
- (Crosby, 1979)
- (Nada, 2006)

Contacts

sebastian.giron@ucuenca.edu.ec
juan.jimenezf@ucuenca.edu.ec

